

SYSTEM

A set of detailed methods, procedures and routines created to carry out a specific activity, perform a duty, or solve a problem.

PROCESS

A process is a coordinated and standardized flow of activities performed by people or machines, which can traverse functional or departmental boundaries to achieve business objectives and creates value for internal or external customers.

TYPES OF DATA

MASTER DATA

Relatively permanent data & not expected to change frequently.

Types:-

- ① Accounting Master Data.
- ② Inventory Master Data
- ③ Payroll Master Data
- ④ Statutory Master Data

NON-MASTER DATA

Non permanent data and expected to change frequently.

Also called transaction data.

IMPORTANT DIFFERENCE!

[4m] [2m]

- ① Master data - not typed; selected from available list.
- ② Master data entry - less frequently updated.
- ③ Non Master data - typed; not selected, keeps on changing.
- ④ Master data - selected to maintain standardization
- ⑤ Master data - selected to avoid confusion.

VOUCHER NUMBER

Unique identity of any voucher/document is its VOUCHER NUMBER.

- Voucher Number must be unique
- Separate Numbering Series
- Can have prefix/suffix/ or both.
- must be numbered serially.
- Vouchers must be recorded in a chronological order.

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COMPUTER RELATED OFFENCES

SECTION 43

[COMPUTER में गतिविधि]

- Access to computer
- Downloads, copies or extracts data
- Introduces computer contaminant or computer virus
- Damages computer
- Disrupts
- Denies or causes denial of access
- Tampering/ manipulating
- Destroys, Deletes, Alters Info.
- Steals, Conceals, Destroys or Alters

Penalty under Section 66:

imprisonment - upto 3 years
fine - 5 lakh rupees.

SECTION 65

Tampering with Computer Source Documents

↓
Computer Source Code

SECTION 66E

Punishment for violation of Privacy

Penalty:-

imprisonment - upto 3 years

fine - 2 lakh rupees

1) Section 66B

Punishment for dishonestly receiving stolen computer resource or communication device. ★

2) Section 66C

Punishment for identity theft. ★

3) Section 66D

Punishment for cheating by personation by using computer resource. ★

★ Penalty:-

imprisonment - upto 3 years
fine - ₹ 1 lakh.

SPDI

- Banks are one of largest stakeholders of SPDI.
- Every bank should communicate, develop and host privacy policy of bank.
- The policy should include all key aspects of how they deal with the personal information collected by the bank.

PRIVACY POLICY

- It is mandatory to ensure security of personal information.
- Information should be protected using appropriate security standards such as ISO 27001.
- Information collected is to be confirmed with the customers.
- The customer should be able to control how their non personal information is collected and used online.

STAGES OF MONEY LAUNDERING

① PLACEMENT:

movement of proceeds, frequently currency, from the scene of the crime to a place, or into a form, less suspicious and more convenient of the criminal.

② LAYERING:

Layering involves the repetition of proceeds from illegal source using complex transactions designed to obscure the audit trail and hide the proceeds.

③ INTEGRATION:

Integration involves conversion of illegal proceeds into apparently legitimate business earnings through normal financial or commercial operations.

ANTI-MONEY LAUNDERING USING TECHNOLOGY

- CBS transactions are integrated & there is a greater risk.
- Banks are using special fraud & risk management software to prevent and detect fraud and integrate this as part of their internal process and daily processing and reporting.

FINANCING OF TERRORISM

- funds are moved via wire transfers and in and out of business & personal accounts
- Illegitimate charities, etc.
- The money frequently starts out clean i.e. as a 'charitable donation' before moving to terrorist accounts.
- It is highly time sensitive requiring quick response

CYBER CRIME

offences that are committed against individuals or groups of individuals with a criminal motive to intentionally harm the reputation of the victim or cause physical or mental harm, or loss, to the victim directly or indirectly, using modern telecommunication networks such as Internet and mobile phones

CLASSIFICATION:

- 1) Unauthorized Access to computers by tracking into systems or stealing passwords
- 2) Computer forgery - involves changing images or data stored in computers.
- 3) Committing of a fraud by Manipulation of the input, output or throughput of a computer based system.
- 4) Deliberate damage caused to computer data or programs through virus programs or logic bombs.
- 5) Unauthorized reproduction of computer programs or software piracy.

Access For Many Damages Reproduction

APPLICATION SOFTWARE

4 GATEWAYS OF APPLICATION CONTROL

Configuration

- Defining access rules from various devices / terminals
- Creation of user types
- Creation of Customer Type, Deposit Type, Year-end process
- User Access & Privileges
- Password Management

Masters

- Customer Master for Advances
- Deposit Master
- Customer Master
- Employee Master
- Income Tax Master

Transactions

- Deposit Transactions
- Advances Transactions
- ECS Transactions
- General Ledger

Reports

- Summary of transactions a day
- Daily General Ledger (GL)
- Activity Logging Reviewing
- MIS Report for each product or service
- Reports covering performance, compliance
- Reports of Exception

RISK PREDICTION as per BASEL III using ARTIFICIAL INT.

- complex business relationships and risks cannot be quantified statistically.
- Traditional MIS do not address answers to hard questions on data.
- Artificial Neural Network Logic (AI) → algorithms based on neural networks are executed on data warehouse, so as to understand hidden trends for risk assessment.
- Helps in assessment of capital adequacy under Basel.

MONEY LAUNDERING

[4m]

Short Note

- process by which proceeds of crime are concealed or made opaque to come from a legitimate source.
- Objective to conceal the existence, illegal source, or illegal application of income.
- Dirty (tainted) money appear clean (untainted)

Definition

Sec 3 of PMLA
Whoever directly or indirectly attempts to indulge or knowingly assists or knowingly is a party or is actually involved in any process or activity connected with the proceeds of crime and projecting it as untainted property shall be guilty of the offence of money-laundering.

Key Aspects of PMLA

- Maintenance of record of all cash transactions above ₹ 10 lakhs
 - All series of cash transactions of value less than ₹ 10 lakhs within a month (aggregate value above ₹ 10 lakhs)
 - All cash transactions have forged or counterfeit notes have been used
 - All suspicious transactions made in cash or otherwise
- Penalty for Non compliance shall not be less than 10 thousand but which may extend to ₹ 100000 for each such failure.

INFORMATION SECURITY

Sub Processes

- 1) **Information Security Policies, Procedures and Practices**
Approval / Implementation of security policy.
- 2) **User Security Administration**
Complete administration of user rights from creation to disabling of users.
- 3) **Application Security**
How security is implemented at various aspects.
- 4) **Physical Security**
Refers to security implemented through physical access controls.
- 5) **Database Security**
Security over Database Software.
- 6) **Operating System Security**
Security of operating systems installed in the servers.
- 7) **Network Security**
Refers to how security is provided at various layers of network.

Risks & Controls

RISKS	CONTROLS
① Lack of Management Direction and Commitment to Protect Information Assets	Security Policies are established and management monitors compliance with policies.
② User Accountability is not established	All users are required to have a unique User ID.
③ Security Breaches may go undetected	Access to sensitive data is logged and the logs are regularly reviewed by management.
④ Potential loss of data integrity, Confidentiality, availability	Physical access restrictions are implemented and administered.

APPLICATION SOFTWARE

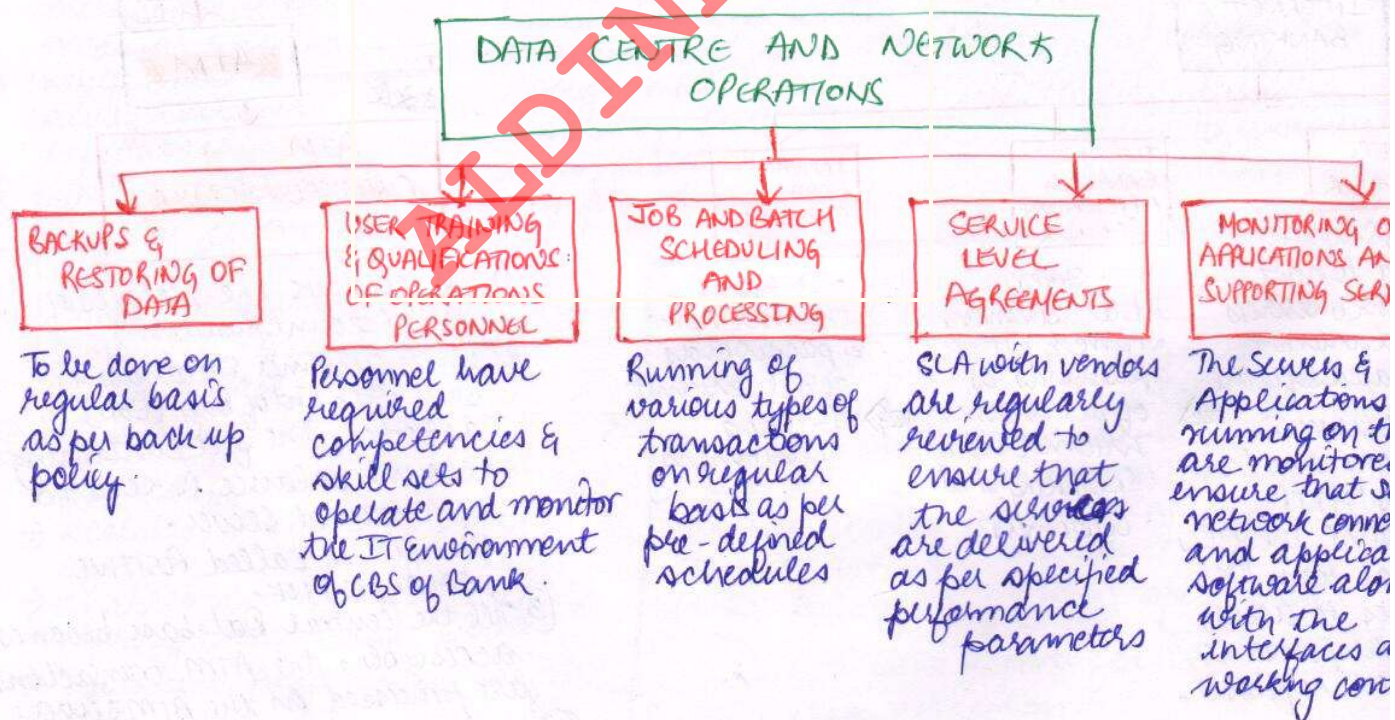
RISKS	CONTROLS
① Interest may be incorrectly computed leading to incorrect recording of income/expense	Interest is automatically & correctly computed, rounded off and appropriately accrued.
② Inappropriate reversal of charges resulting in loss of revenue	System does not permit reversal of the charges in excess of the original amount charged.
③ Inappropriate set up of accounts resulting in violation of business rules.	The system parameters are set up as per business process rules of the bank.
④ Incorrect classification and provisioning of NPAs, resulting in financial misstatement.	Configuration/customization exists in the application to perform the NPA classification as per relevant RBI guidelines.

RISKS AND CONTROLS FOR VARIOUS BANKING PROCESSES

[Refer Pg No 43 to 48]

RISKS ASSOCIATED WITH CBS

- ① Access Controls : Designing, monitoring is difficult
- ② User Identity Management : more than 5000 users may interact at one
- ③ Authentication Procedures : may be inadequate & user may not be traceable
- ④ Authorization Procedures : Data can be entered directly.
- ⑤ Maintaining Response Time : Ensuring optimum response time can be challenging.
- ⑥ Incident Handling Procedures : May not be adequate
- ⑦ Several software Interfaces : Can have 75-100 different interfaces across diverse networks
- ⑧ Change Management : At application level and data level
- ⑨ Ownership of Data/Process : Data resides at the Data Center - Establish clear ownership. [Access (4) RISC Ownership]



Back To Job and Service on Monday

CORE BANKING SYSTEM

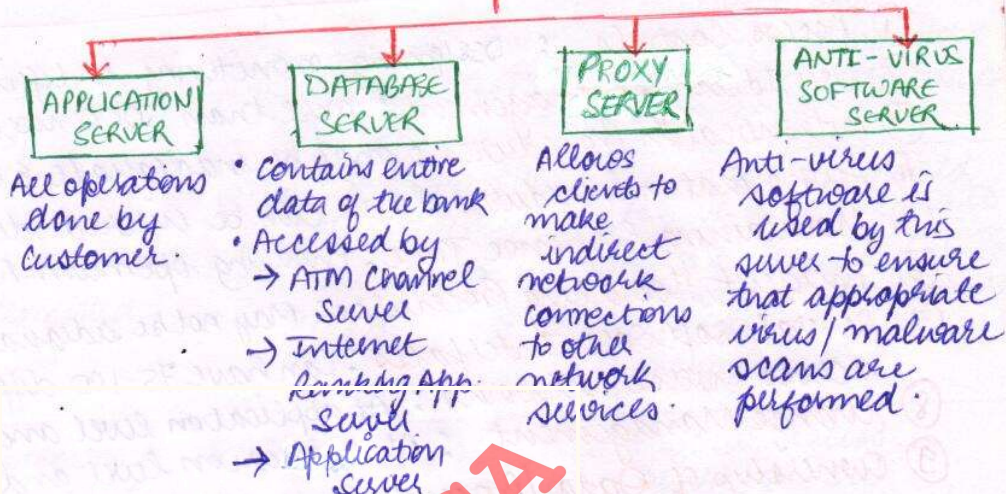
FUNCTIONAL ARCHITECTURE

How does CBS Work?

- Planning
- Approval
- Selection
- Design and Development
- Procured
- Testing
- Implementation
- Maintenance
- Support
- Updation
- Audit

TECHNOLOGICAL ARCHITECTURE

(I) GENERAL:



(II)

INTERNET BANKING

WEB SERVER

- Web Server is used to host all web services and internet related software.
- A web server is a program that uses HTTP to serve the files that form web pages to users, in response to their requests.

INTERNET BANKING APPLICATION SERVER

- It contains Internet Banking Software. It authenticates the customer with the login details stored in IBCS.

INTERNET BANKING CHANNEL SERVER

- It stores the user name & passwords of all internet banking customers.

(III)

ATM

ATM CHANNEL SERVER

- ① It contains the details of ATM Account Holders.
- ② When customer Database is busy with End of Day (EoD) activities, the file containing account balance is sent to ATM channel server. This file is called POSITIVE BALANCE FILE.
- ③ Till the central Database becomes accessible, the ATM transactions are processed on the ATM server.
- ④ Once the central database server becomes accessible, all the transactions that took place till such time as the central database became inaccessible would be updated in the central database.

CORE BANKING SYSTEM

(2 or 4m)

- Centralized On line Real-Time Environment
- central shared database supports the entire banking application.
- update a common database located at Data Center
- Branches Function only as Delivery Channels.

EXAMPLES

① FINNACLE: (INFOSYS)

- developed by Infosys
- universal banking functionality
- covers all modules covering all services.

② FINNONE: (NUCLEUS)

- web based Global Banking Product
- supports banks & financial solⁿ comp.

③ FLEXCUBE: (ORACLE)

- Comprehensive, integrated, interoperable and modular solution
- enables banks to manage evolving customer expectations.

④ BANCS: (TCS)

- customer centric business model
- simplified operations comprising loans, deposits, wealth management, etc.

⑤ bankMate: (IBM)

- full scale banking solution
- scalable, integrated e-banking systems
- enables communication through any touch point.
- Anywhere, Anytime paradigm

ARCHITECTURE

[FUNCTIONAL]

CORE FEATURES

① INFORMATION FLOW:-

streamline integration and unite corporate information.

② CUSTOMER CENTRIC:-

right offers at the right time with the right channel.

③ REGULATORY COMPLIANCE:-

has built in and regularly updated regulatory platform.

④ RESOURCE OPTIMIZATION:-

optimizes utilization of information and resources of banks and lowers costs through improved asset reusability, faster turnaround times, faster processing & improved accuracy.

① Online Real time Processing

② Transactions are posted immediately.

③ All databases update simultaneously.

④ Centralized operations

⑤ Highly dependable system based control

⑥ Increased access by staff at various levels based on authorization

⑦ Daily, Half Yearly and Annual closing.

⑧ Anytime, Anywhere Access to customer and vendors.

FUNCTIONS OF THE BRANCH

- Creating manual documents capturing data required for input into software.
- Internal Authorization
- Initiating Beginning-Of-Day (BOD) operations.
- End-Of-Day (EOD) operations
- Reviewing Reports for control and error correction.

APPLYING IT CONTROLS

GENERAL CONTROLS

APPLICATION CONTROLS

also called
"INFRASTRUCTURE CONTROLS"

- Macro in Nature
- Impact Persuades the IT environment at different layers.

INCLUDES:

- 1) Information Security Policy:
Security policy approved by the senior management.
- 2) Administration, Access & Authentication:
IT should be administered with appropriate policies and procedures.
- 3) Separation of Key IT Functions:
Key demarcation of duties for different personnel within IT department to ensure that there are no SOP conflicts.
- 4) Change Management:
IT solutions and its components must be changed in tune with changing needs.
- 5) Backup, Recovery & Business Continuity:
Resilience of Banking Operations should be ensured by having appropriate business continuity including backup, recovery & off-site data centre.
- 6) Management of Systems Acquisition & Implementation:
Process of acquisition and implementation of systems should be properly controlled.

controls which are implemented in an application to prevent or detect and correct errors. These controls are IN-BUILT in the application software to ensure accurate and reliable processing.

EXAMPLES:

- 1) Data Edits: allowed only for permissible fields.
- 2) Exception Reporting: all exceptions should be reported.
- 3) Balancing of Processing Totals: debit and credit of all transactions are tallied.
- 4) Error Reporting: Errors in processing are reported.
- 5) Separation of Business Functions: Transaction Initiation vs Authorization
- 6) Transaction Logging: All transactions are identified with unique ID and logged.

IAS
Change
Black Money